

Infinite Canvas Whiteboard

OJT Coding Exam — Acceptance Criteria

Adaptive Hub Exam | Stack: Laravel + Konva.js + DB of Choice | Time Limit: 4 Hours

Overview

The OJT candidate must build a Draw.io-inspired infinite whiteboard web application using Laravel as the backend and Konva.js as the canvas library. Users can draw freehand, add shapes (rectangle, circle, line, arrow), add text labels, pan and zoom the canvas freely, and save their work to the database. Saved boards can be reloaded, renamed, and deleted. No user authentication is required — boards are identified by a unique name. The candidate may use any database they are comfortable with (MySQL, PostgreSQL, SQLite) as long as it is supported by Laravel.

Reference System

The expected output is similar to Draw.io (diagrams.net) in terms of core interactions:

- Infinite canvas that can be panned by dragging the background and zoomed with the mouse wheel
- Toolbar with tools: Select, Freehand Draw, Rectangle, Circle, Line, Arrow, Text
- Selected shapes can be moved, resized, and deleted
- Save button persists the entire canvas state to the database as JSON
- Boards list page where saved boards can be opened, renamed, or deleted

System Architecture

- Backend: Laravel (latest stable) running on Laragon — serves API and Blade views
- Database: Candidate's choice — MySQL, PostgreSQL, or SQLite — must be supported by Laravel
- Canvas Library: Konva.js (loaded via CDN) — handles all drawing, shapes, pan, zoom
- Frontend: Blade templates with inline JavaScript — no separate frontend framework required
- Canvas state saved as JSON string in the database — Konva.js `toJSON()` and `loadJSON()` used

1. Infinite Canvas & Navigation

#	Acceptance Criteria	Priority	Evaluation Method
AC-01	The canvas renders as a full-screen infinite whiteboard using Konva.js Stage and Layer	Must Have	Open app — full screen canvas visible
AC-02	User can pan the canvas by clicking and dragging the background (not a shape)	Must Have	Drag background — canvas moves
AC-03	User can zoom in and out using the mouse scroll wheel — minimum zoom 20%, maximum 300%	Must Have	Scroll up/down — canvas zooms in/out
AC-04	A zoom percentage indicator is shown on screen (e.g. 100%)	Should Have	Visual inspection — zoom % visible
AC-05	A reset view button recenters and resets zoom to 100%	Should Have	Click reset — canvas returns to default view

2. Toolbar & Drawing Tools

#	Acceptance Criteria	Priority	Evaluation Method
AC-06	A visible toolbar shows the following tools: Select, Freehand, Rectangle, Circle, Line, Arrow, Text	Must Have	Visual inspection of toolbar
AC-07	The active tool is visually highlighted in the toolbar	Must Have	Click each tool — active state visible
AC-08	Freehand tool draws a continuous stroke on the canvas as the user drags the mouse	Must Have	Select freehand — draw on canvas
AC-09	Rectangle tool draws a rectangle by click-drag — shows preview while dragging	Must Have	Select rectangle — drag on canvas
AC-10	Circle tool draws a circle/ellipse by click-drag — shows preview while dragging	Must Have	Select circle — drag on canvas
AC-11	Line tool draws a straight line between two points (click start, click end)	Must Have	Select line — click two points
AC-12	Arrow tool draws a line with an arrowhead at the end point	Must Have	Select arrow — click two points — arrowhead visible
AC-13	Text tool places an editable text label on the canvas at the clicked position	Must Have	Select text — click canvas — type label
AC-14	A color picker allows the user to change the stroke/fill color before drawing	Must Have	Change color — draw shape — confirm color applied
AC-15	A stroke width selector (thin, medium, thick) controls line thickness	Should Have	Change thickness — draw — confirm applied

3. Shape Selection & Editing

#	Acceptance Criteria	Priority	Evaluation Method
AC-1 6	Select tool allows clicking a shape to select it — selected shape shows handles/border	Must Have	Click shape with Select tool — handles appear
AC-1 7	Selected shape can be dragged to a new position on the canvas	Must Have	Select shape — drag — shape moves
AC-1 8	Selected shape can be resized by dragging its corner handles	Must Have	Select shape — drag corner — shape resizes
AC-1 9	Pressing the Delete or Backspace key removes the selected shape from the canvas	Must Have	Select shape — press Delete — shape removed
AC-2 0	Clicking the canvas background deselects the current shape	Should Have	Click background — handles disappear
AC-2 1	Double-clicking a text label opens it for editing in place	Should Have	Double-click text — edit inline

4. Save & Load Canvas

#	Acceptance Criteria	Priority	Evaluation Method
AC-2 2	A Save button serializes the entire Konva.js canvas state to JSON using stage.toJSON()	Must Have	Click Save — POST request sent to Laravel API
AC-2 3	The JSON canvas state is saved to the boards table in the database	Must Have	Inspect boards table — canvas_data column populated
AC-2 4	On page load, if a board exists in the DB, the canvas is restored using Konva.Stage.create(json)	Must Have	Save board — refresh page — canvas reloaded correctly
AC-2 5	A success notification or message is shown after a successful save	Must Have	Click Save — success message appears
AC-2 6	If save fails, a user-friendly error message is displayed — page does not crash	Must Have	Simulate API failure — error message shown
AC-2 7	Auto-save triggers every 60 seconds if the canvas has been modified	Should Have	Modify canvas — wait 60 sec — check DB updated

5. Board Management — CRUD

#	Acceptance Criteria	Priority	Evaluation Method
AC-28	A boards list page shows all saved boards with name and last updated date	Must Have	Visit /boards — list of boards shown
AC-29	User can create a new blank board by entering a board name	Must Have	Click New Board — enter name — blank canvas opens
AC-30	User can open an existing board from the list — canvas loads its saved state	Must Have	Click board from list — canvas restores correctly
AC-31	User can rename a board from the boards list page	Must Have	Click rename — enter new name — saved in DB
AC-32	User can delete a board — board and its canvas data are removed from the database	Must Have	Click delete — confirm — board removed from list and DB
AC-33	Board names must be unique — duplicate names are rejected with a validation error	Should Have	Create two boards with same name — expect error

6. Version Control

#	Acceptance Criteria	Priority	Evaluation Method
AC-34	Project is initialized as a Git repository	Must Have	Check for .git folder in project root
AC-35	.gitignore excludes: .env, /vendor, /node_modules, /storage	Must Have	Inspect .gitignore — confirm entries present
AC-36	At least 5 meaningful commits made throughout development — not one final commit	Must Have	Run git log — verify commit count and messages
AC-37	Commit messages are descriptive (e.g. feat: add rectangle tool, fix: canvas not saving)	Should Have	Run git log --oneline — review message quality
AC-38	.env.example committed with all required keys but no real values	Must Have	Check repo — .env.example present, .env absent

7. Database & Migrations

#	Acceptance Criteria	Priority	Evaluation Method
AC-39	Candidate declares their chosen database in .env (DB_CONNECTION) and documents it in README	Must Have	Check .env.example and README — DB choice stated
AC-40	Migration exists for the boards table — runs without errors on the chosen DB	Must Have	Run php artisan migrate — no errors
AC-41	boards table has: id, name (unique), canvas_data (longText or equivalent), created_at, updated_at	Must Have	Inspect boards migration file — confirm all columns
AC-42	Board model exists with correct fillable fields: name, canvas_data	Must Have	Code review of Board model
AC-43	Laravel API routes exist for: GET /api/boards, POST /api/boards, GET /api/boards/{id}, PUT /api/boards/{id}, DELETE /api/boards/{id}	Must Have	Check routes/api.php — all 5 routes present

8. UI & Usability

#	Acceptance Criteria	Priority	Evaluation Method
AC-44	The whiteboard UI is clean with toolbar on top or side — canvas takes up the full remaining space	Must Have	Visual inspection
AC-45	Toolbar icons or labels clearly indicate what each tool does	Must Have	Visual inspection — no unlabeled mystery buttons
AC-46	The boards list page is accessible via a navigation link from the whiteboard page	Must Have	Click nav link — boards list opens
AC-47	The app is functional on Chrome desktop (latest version) — no console errors during normal use	Must Have	Open DevTools — use app — confirm no errors
AC-48	A first-time user can figure out how to draw and save without any instructions	Should Have	Evaluator usability test

Scoring Rubric

Category	Weight	Pass Threshold
Canvas & Navigation (AC-01 to AC-05)	10%	AC-01, AC-02, AC-03 passed
Toolbar & Drawing Tools (AC-06 to AC-15)	25%	AC-06 to AC-13 all passed
Shape Selection & Editing (AC-16 to AC-21)	15%	AC-16, AC-17, AC-18, AC-19 passed
Save & Load Canvas (AC-22 to AC-27)	20%	AC-22, AC-23, AC-24, AC-25 passed
Board CRUD (AC-28 to AC-33)	15%	AC-28, AC-29, AC-30, AC-31, AC-32 passed

Category	Weight	Pass Threshold
Version Control (AC-34 to AC-38)	5%	AC-34, AC-35, AC-36, AC-38 passed
Database & API (AC-39 to AC-43)	5%	AC-40, AC-41, AC-43 passed
UI & Usability (AC-44 to AC-48)	5%	AC-44, AC-45, AC-46, AC-47 passed
TOTAL	100%	70% or higher to pass

Overall passing score: 70% or higher. Must Have criteria in AC-01, AC-02, AC-03, AC-08, AC-09, AC-22, AC-23, AC-24, AC-28, AC-36, AC-40, and AC-43 must each be individually met regardless of total score.

Estimated Completion Time (4-Hour Target)

Module	Task Summary	Est. Time
Project Setup	Laragon, Laravel install, .env, DB config, Konva.js via CDN	20 min
Git Setup	git init, .gitignore, initial commit, .env.example	10 min
Database & API	boards migration, Board model, 5 API routes in routes/api.php	25 min
Boards List Page	Blade view — list boards, new board form, rename, delete buttons	30 min
Canvas Setup	Konva.js Stage + Layer, full screen, pan + zoom with mouse wheel	35 min
Drawing Tools	Toolbar UI, freehand, rectangle, circle, line, arrow, text tools	55 min
Select & Edit	Select tool, drag to move, resize handles, delete key, deselect	30 min
Save & Load	stage.toJSON() on save, Konva.Stage.create() on load, API calls	25 min
UI Polish & Testing	Toolbar labels, nav link, color picker, end-to-end test, commits	30 min
TOTAL		4 hrs

The drawing tools section is the most time-consuming step — the candidate should implement them one by one starting with Rectangle (easiest) then Freehand, Circle, Line, Arrow, and Text. Save & Load must be tested early to avoid last-minute surprises with Konva.js JSON serialization.

Exam Notes & Deliverables

- Time limit: 4 hours
- Environment: Laravel running on Laragon (local) — no deployment required
- Canvas library: Konva.js loaded via CDN — no npm install required
- Frontend: Blade templates with inline JavaScript — no Vue, React, or separate SPA

- Database: Candidate's choice — MySQL, PostgreSQL, or SQLite — must be Laravel-supported and documented in README
- Candidate must declare DB_CONNECTION in .env.example so evaluator can set up correctly
- Canvas state must be saved as Konva.js JSON (stage.toJSON()) — no custom format
- Version control: Git required — minimum 5 commits with descriptive messages
- .env must NOT be committed — only .env.example with blank values
- Candidate must submit: full source code, .env.example, and a README with setup steps
- Evaluator will run php artisan migrate and php artisan serve to test the submission
- Reference: Candidate may look at Draw.io (diagrams.net) as UI inspiration — not as code
- The reference implementation must not be shared with the candidate before the exam

Prepared by Adaptive Hub Team